

Stress-Testing Autonomous Vehicles and Beyond: Mcity and Infrastructure Resilience

Henry Liu

Director, UMTRI & Mcity

Bruce Greenshields Professor of Engineering

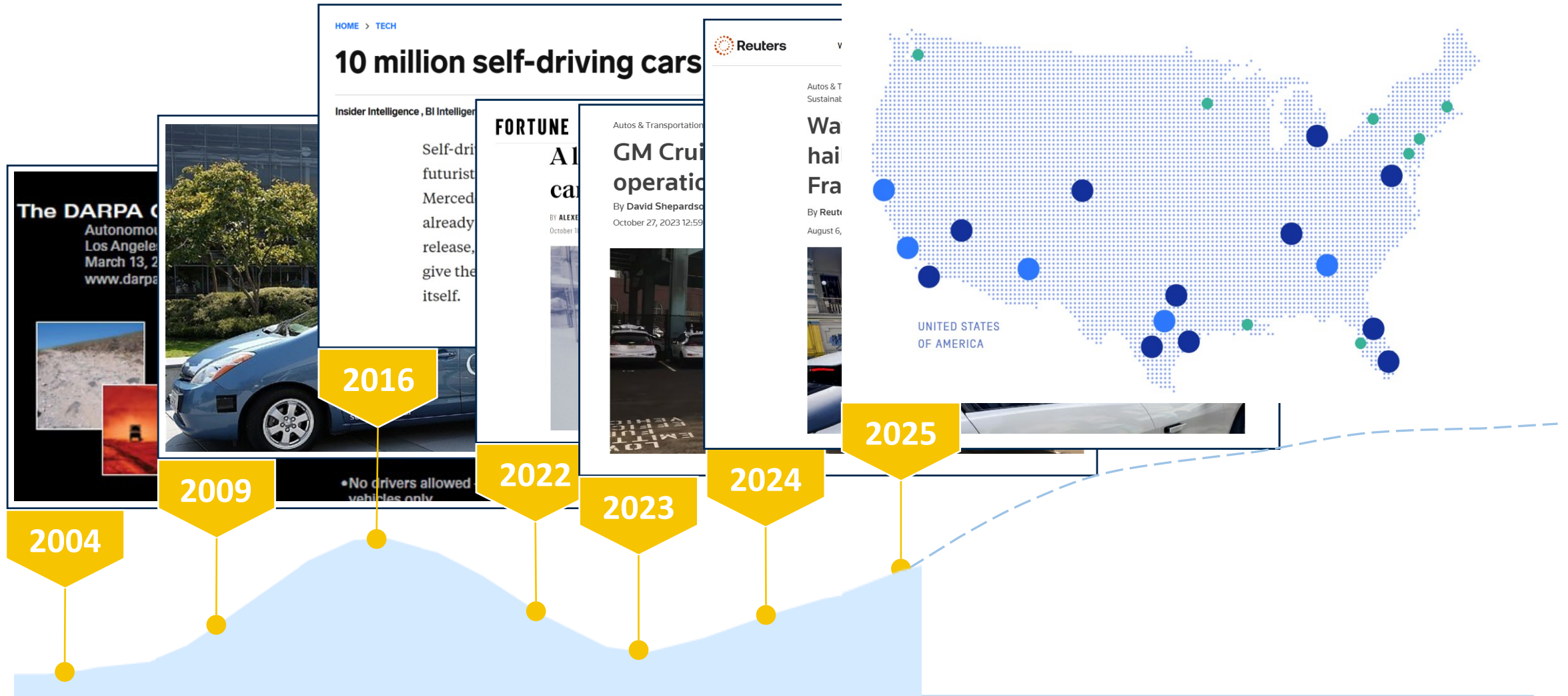
University of Michigan, Ann Arbor

September 17th, 2026

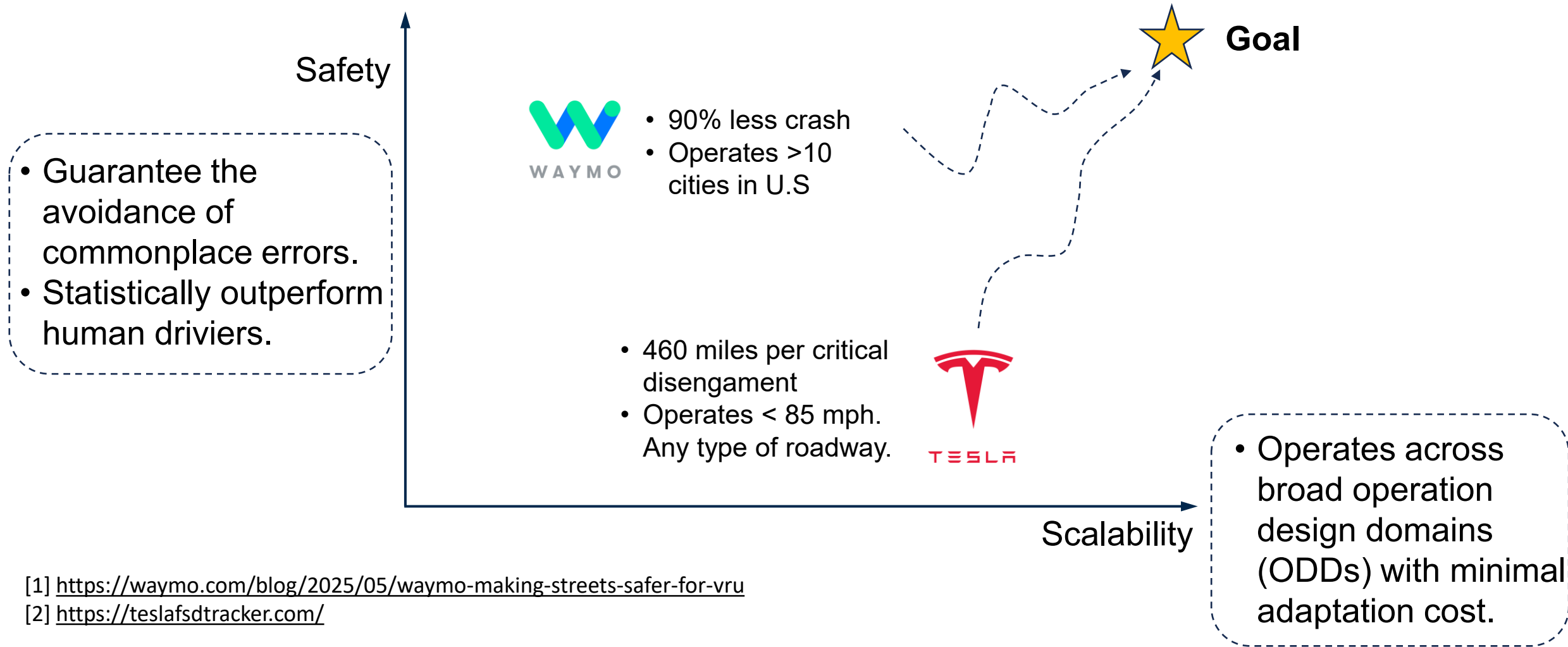
NSF Workshop on Resilience of Cyber-Physical Infrastructure Systems



Autonomous Vehicle Technology: from 2004 to 2026



Safety-Scalability Dilemma



[1] <https://waymo.com/blog/2025/05/waymo-making-streets-safer-for-vru>

[2] <https://teslafdtracker.com/>

Curse of Dimensionality (CoD)

- The CoD Problem is that when the dimensionality increases, the volume of the space increases so fast that the available data become sparse.

Weather



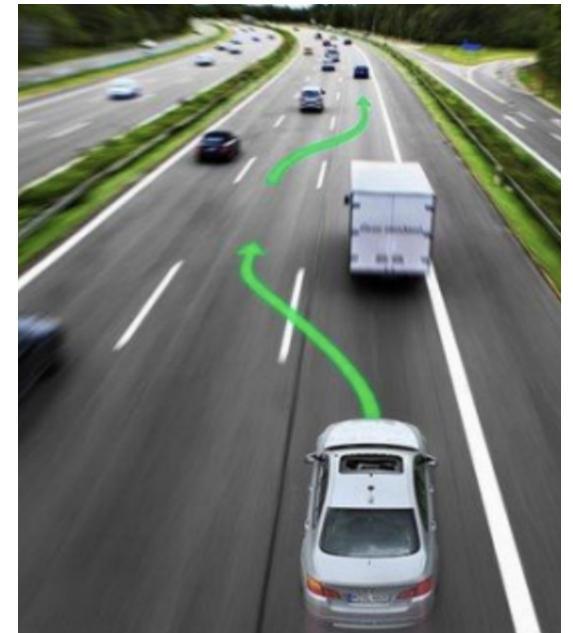
Road Infrastructure



Road Users



Maneuvers

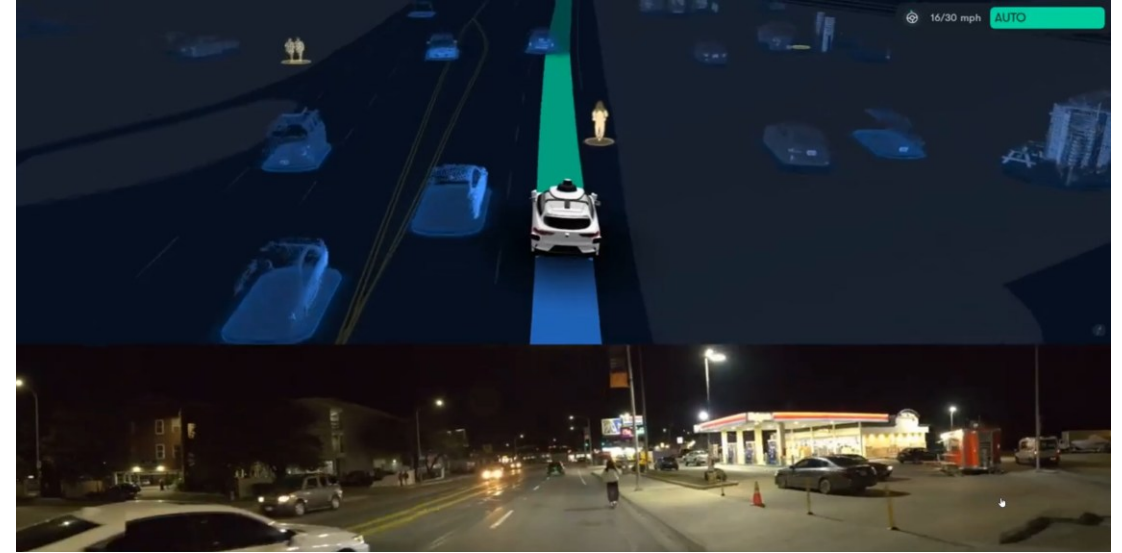


Curse of Rarity (CoR)

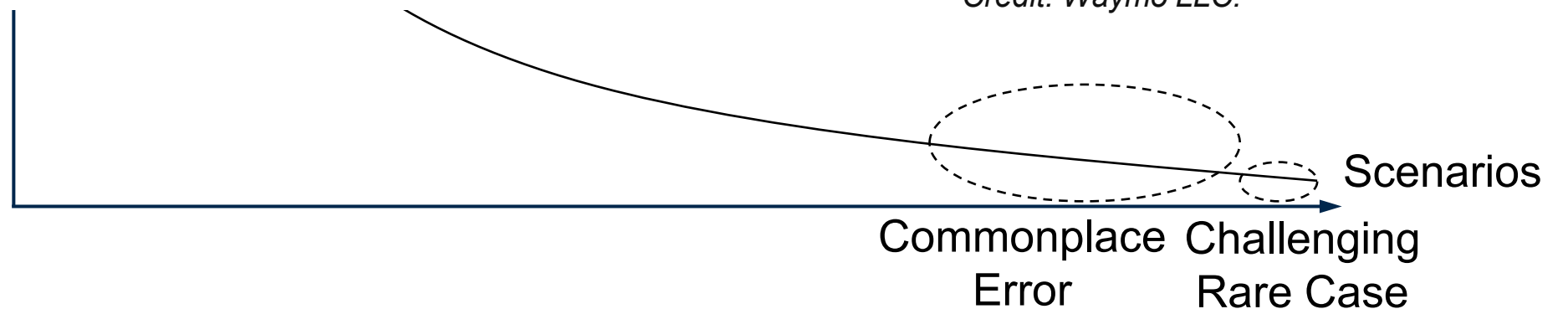
The occurrence probability for the events of interest is so rare that most available data contain very little information.



Credit: KVUE



Credit: Waymo LLC.

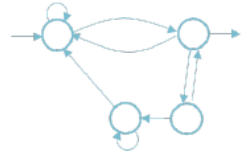


The compounding effects of CoD and CoR pose a structural challenge that creates the safety-scalability Dilemma

- Scaling leads to increased dimensionality, which brings CoD and CoR, then safety suffers
- Improve safety by keeping existing dimensionality, then scaling suffers

Observations on 20 Yrs of AV Development

AV 1.0 by Design
(2004-2017)



Scaling Application

AV 2.0 by Deep Learning
(2012-2024)



Dilemma between Safety and Scalability

AV 3.0 by Foundational Models
(2022 -)



- The safety–scalability dilemma has also led to repeated engineering back-and-forth and technical debates during AV development.
- Increasing dimensionality from AV 1.0 to AV 2.0 and 3.0
 - CoD and CoR
- AV 1.0, 2.0 and 3.0 are not exclusive, many components from prior stages remain

Future Directions

- “How safe is safe enough?” is a central question
 - Benchmarking with human driver, “apple” vs. “apple”
 - Consumer acceptance related to AV capacity boundary
 - City traffic management considering AVs
 - Regulatory framework and market incentive mechanism
- Scaling driving intelligence beyond driving domain
 - Be able to absorb external knowledge without eroding its driving competence
- Generalization of intelligence across different AV platforms
 - Currently not transferrable from one platform to another
 - Mobility foundational model for autonomy transfer

AV Safety Testing

Functional Safety (FuSa)



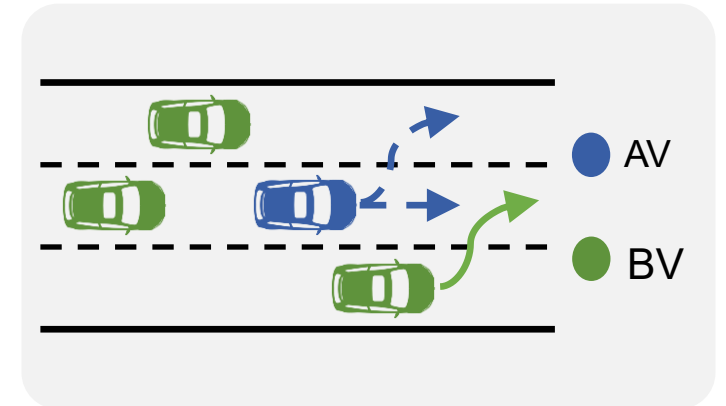
*Hardware/software
Malfunctions
(e.g., ISO 26262)*

Safety Of The Intended Functionality (SOTIF)



*System Insufficiencies
(e.g., ISO 21448)*

Behavior Competency



*System impact at scale
(no standard)*

Behavioral safety testing for autonomous vehicle

- **Basic procedure:** drive AVs in naturalistic driving environment (NDE) for n times and experience m events, and then the estimated rate is m/n .

$$P(A) = \sum_{X \in \mathbb{X}} P(A|X)P(X) \approx \frac{1}{n} \sum_{j=1}^n P(A|X_j) \approx \frac{m}{n}, X_j \sim P(X)$$

A : event of interest (e.g., accident)

X : decision variables of NDE

n : number of tests

m : number of events occurred during the tests

Naturalistic Driving Environment (NDE)

- The decision variables of a **naturalistic driving environment** can be represented as

$$X = \begin{bmatrix} X_{1,1} & \cdots & X_{1,T} \\ \vdots & \ddots & \vdots \\ X_{N,1} & \cdots & X_{N,T} \end{bmatrix}, X \in \mathbb{X}$$

$X_{i,k}$: variables (positions and speed) of the i -th vehicle at the k -th time step

- The simulated NDE should follow the same distribution with the realistic traffic distribution, which is:

$$X \sim P(X)$$

Rare event estimation with importance Sampling

- Importance sampling: construction of a new sampling distribution

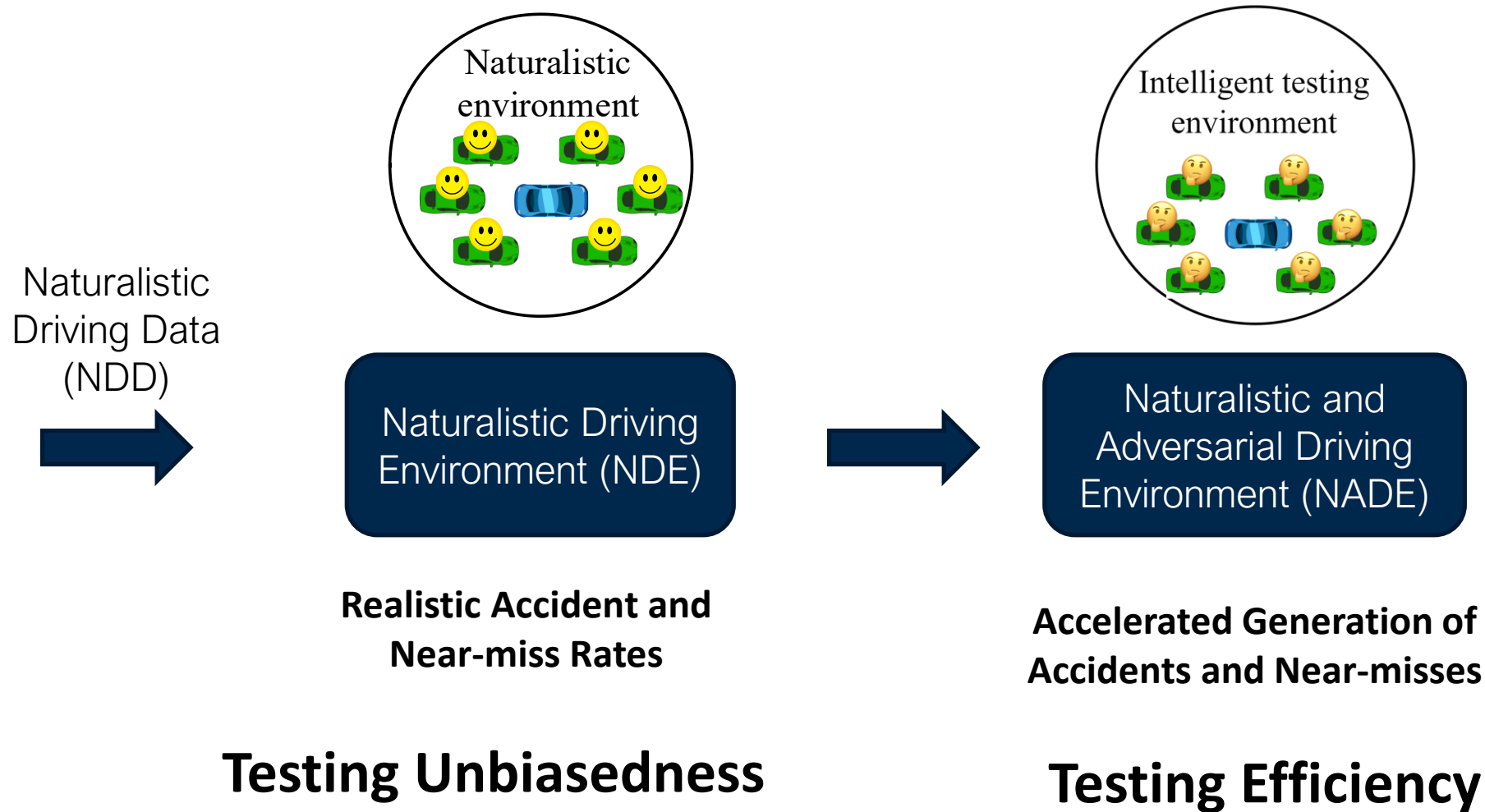
$$P(A) = \sum_{X \in \mathbb{X}} P(A|X)P(X) = \sum_{x \in X} \frac{P(A|X)P(X)}{q(X)} q(X) \approx \frac{1}{n} \sum_{j=1}^n \frac{P(X_j)}{q(X_j)} P(A|X_j), X_j \sim q(X)$$

Construct **an importance function $q(X)$** so that $P(A)$ is an unbiased estimation.

$$q_{\pi}^*(X) \propto \|\nabla_{\pi} P(A_{\pi}|X)\|_2 P(X)$$

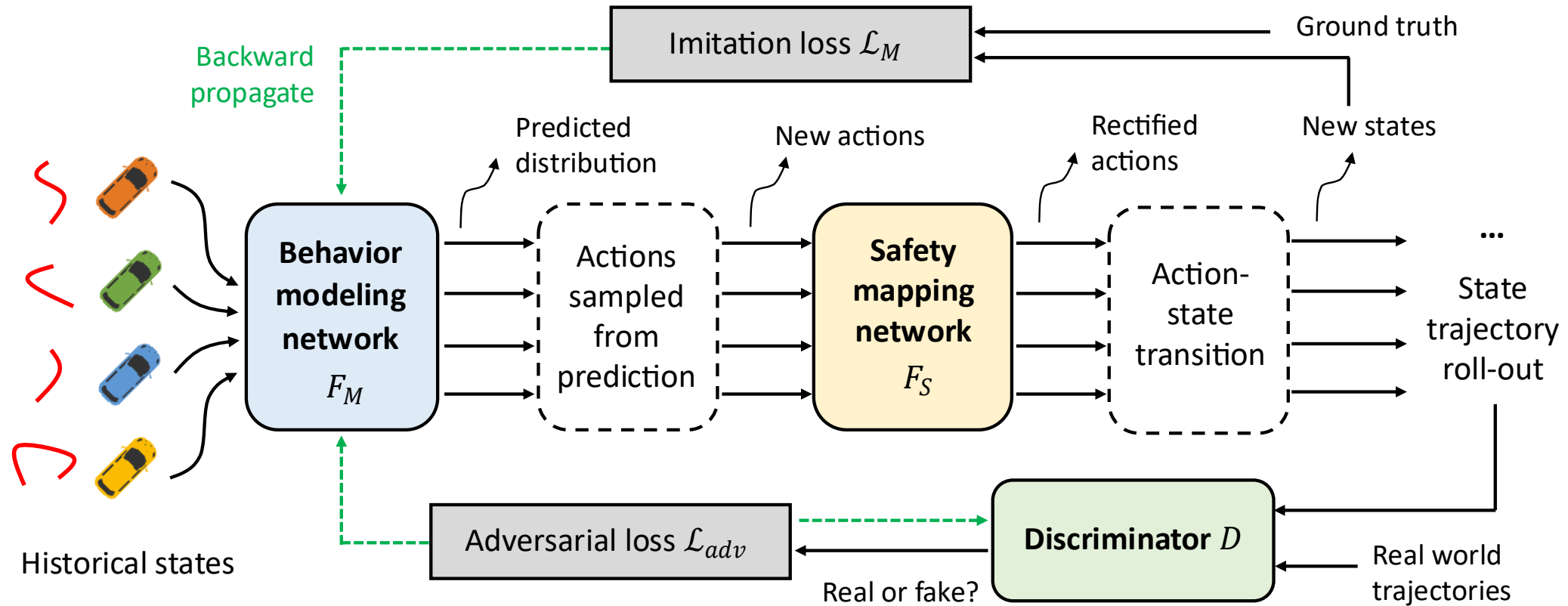
The construction of importance function leads to the development of **naturalistic and adversarial driving environment (NADE)**

Key to the AV Safety Validation Problem

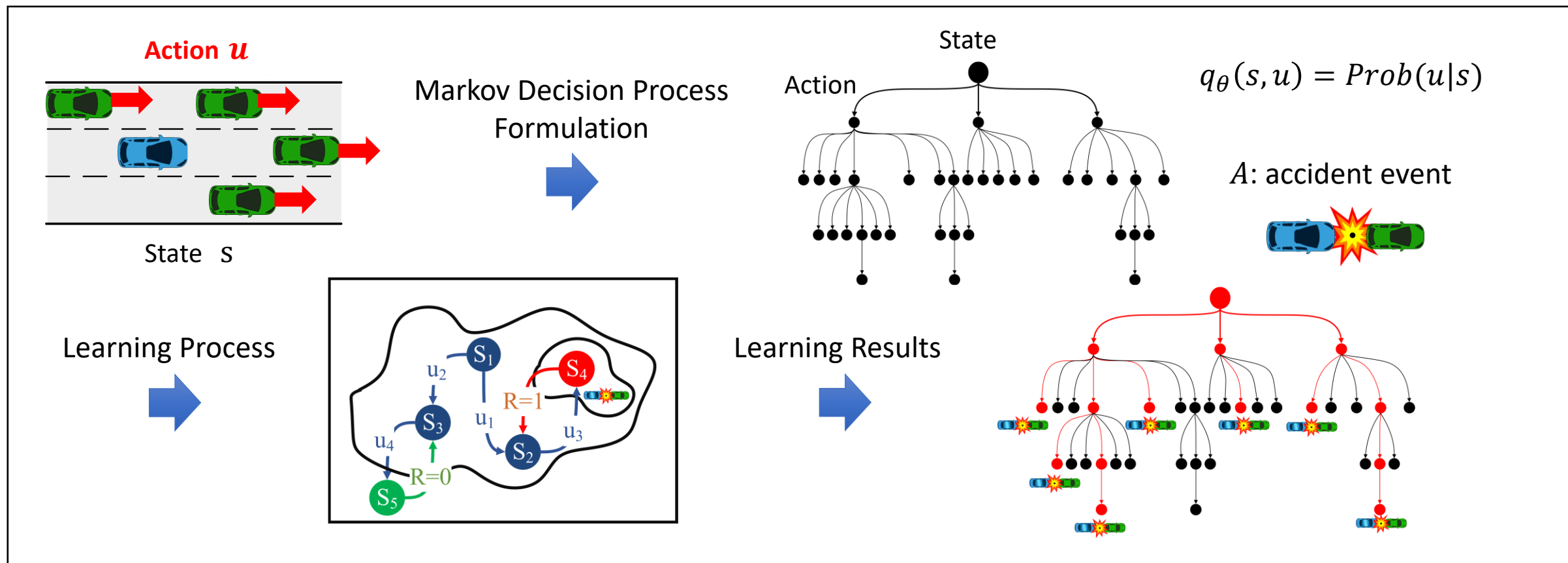


NeuralNDE: A data-driven modeling framework

- Model long-term multi-agent interaction behaviors in complex traffic environment without any heuristic rules.

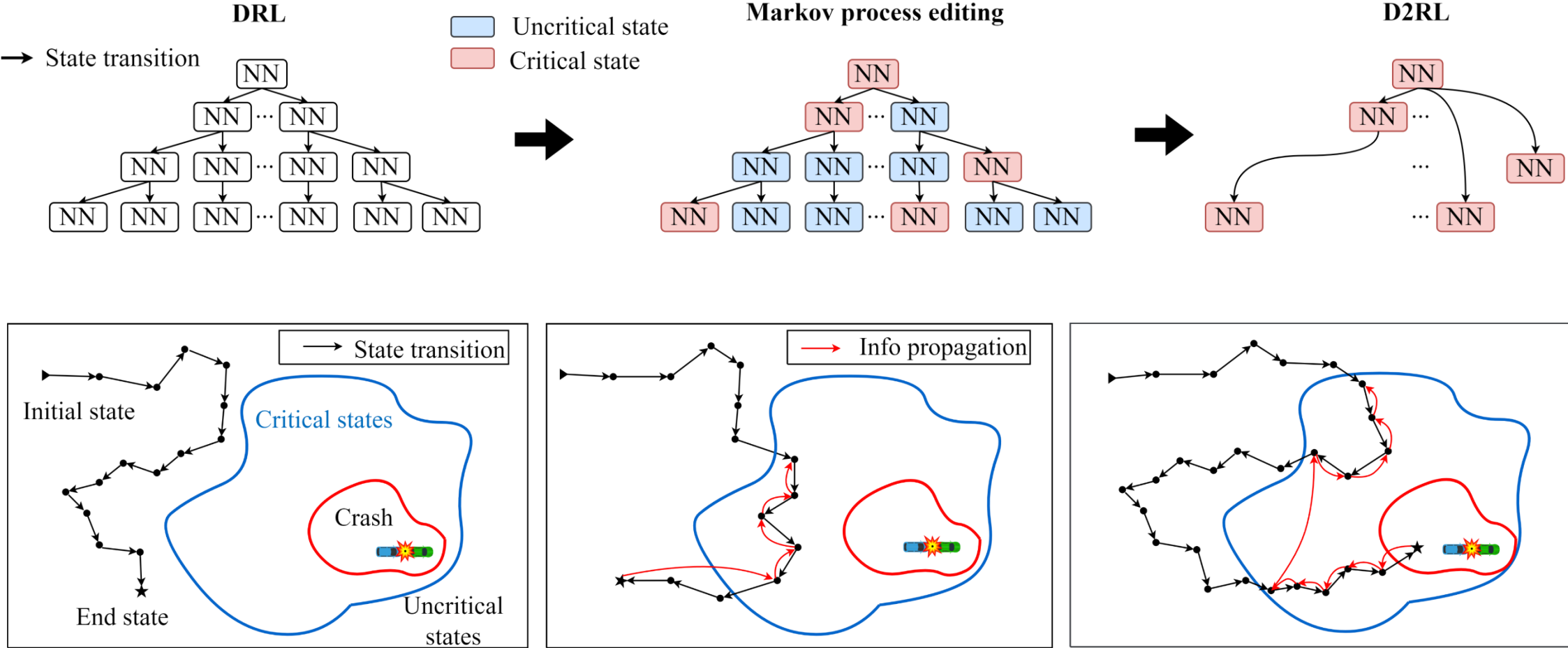


NADE formulated as a Reinforcement Learning Problem



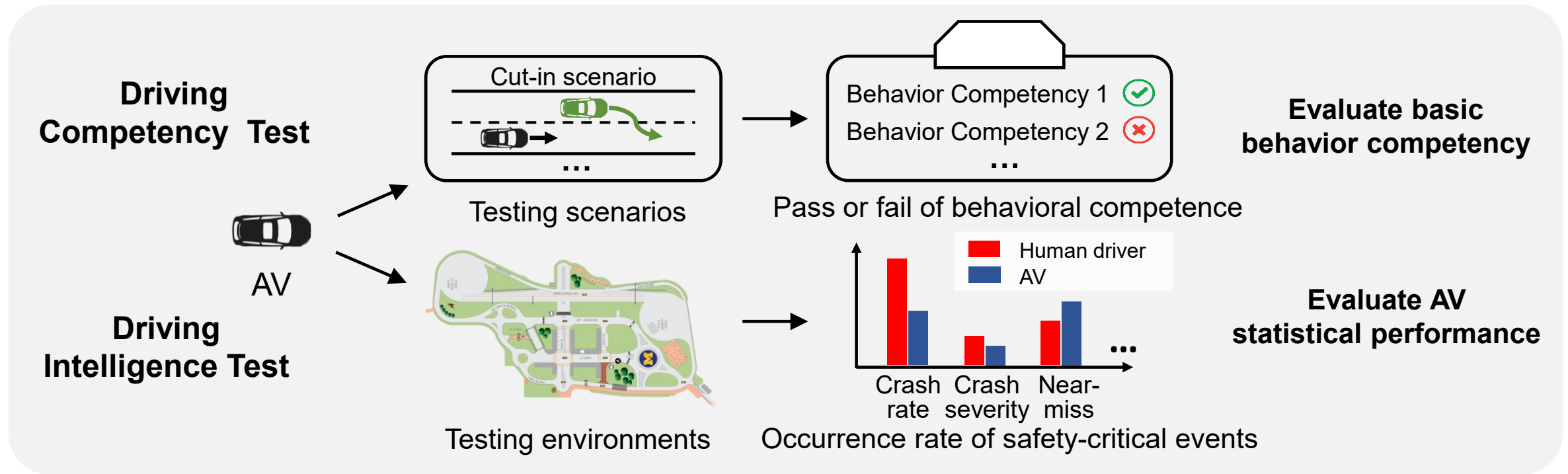
Dense learning algorithms were developed based on the Importance Sampling theory and the Markov Decision Process.

Dense Deep Reinforcement Learning (D2RL)



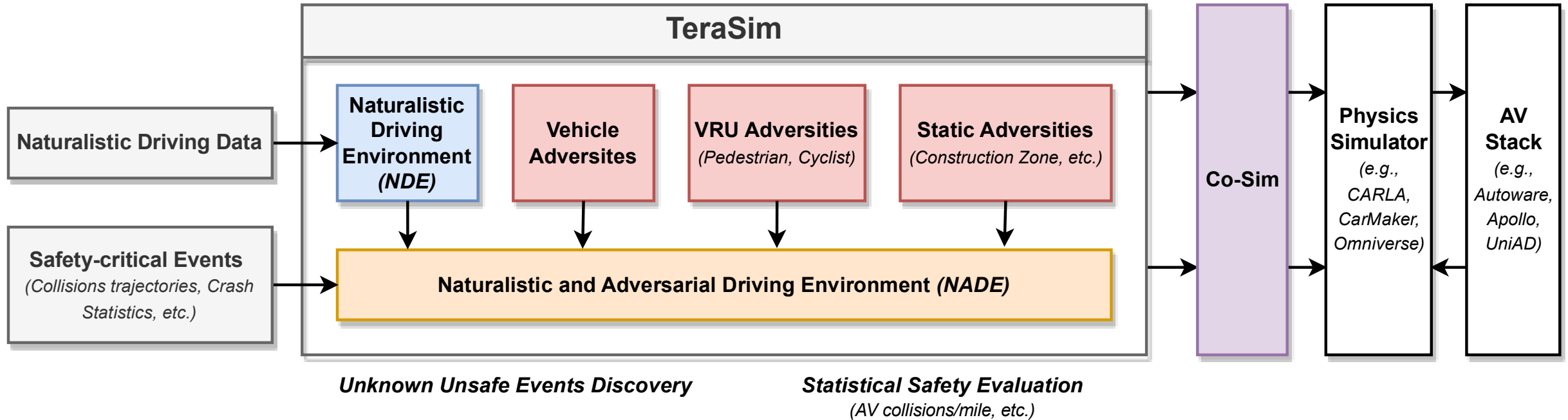
Mcity AV Safety Assessment Program

- Unbiased, rigorous, transparent, and publicly accessible evaluation.



Treats the AV as a **black box** during testing, ensuring fairness and objectivity

TeraSim Architecture



TeraSim is a traffic environment simulator that provides:

- **NDE:** Interactive naturalistic traffic environment learned from real world
- **Adversities:** Diverse stochastic challenging events
- **NADE:** Adversity orchestrator to intelligently challenge the AV
- **Integration:** API for seamless integration with existing simulators

Mcity TeraSim World Model



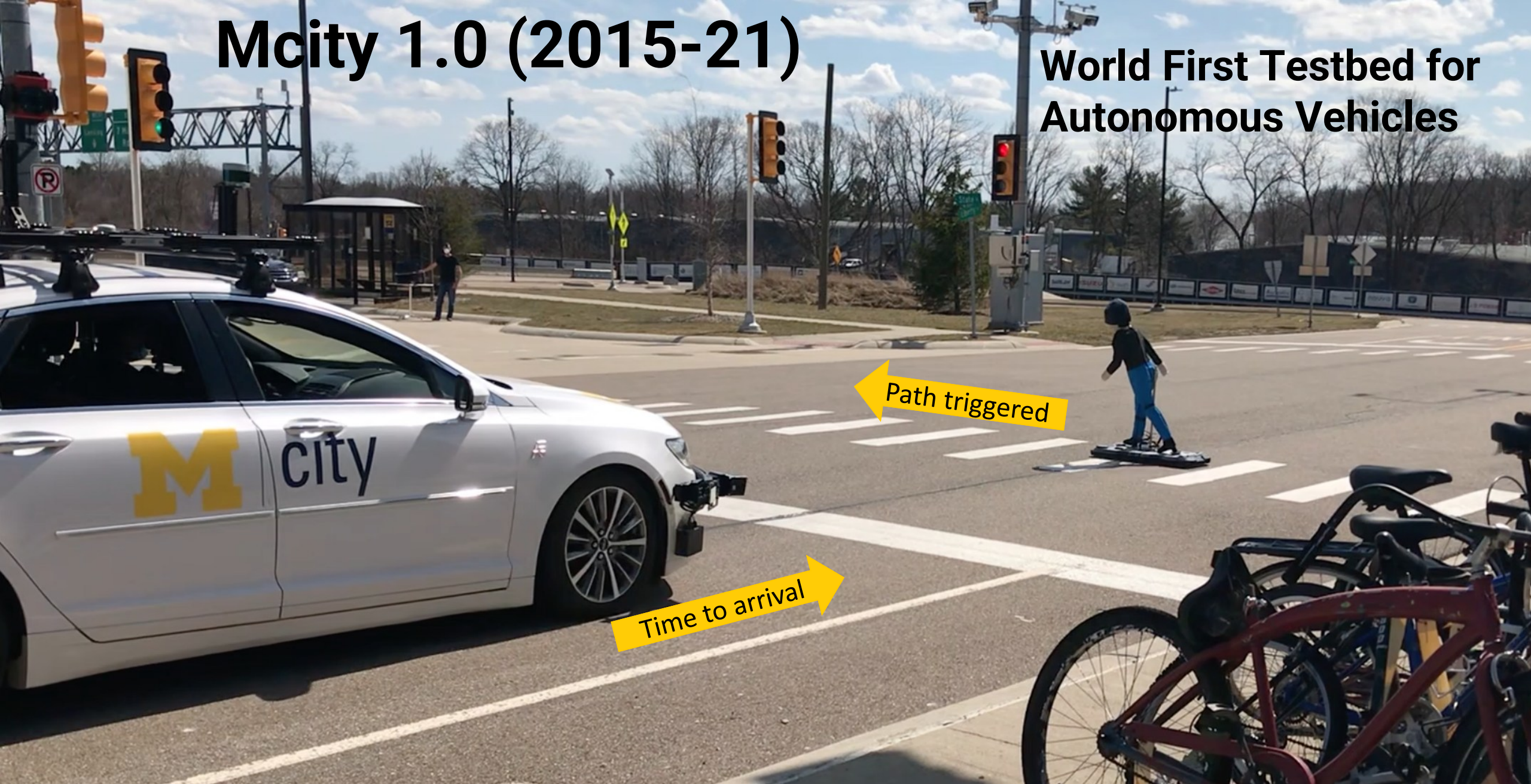
An aerial night photograph of the Mcity test facility. The facility features a complex network of roads with various lane markings, including a large blue circular arrow in the center. The roads are illuminated by streetlights, and some areas have colored lighting (red, green, blue). In the background, a city skyline is visible with a prominent water tower. The text is overlaid in white, bold font.

Mcity test facility is the world's first purpose-built proving ground for connected and automated vehicles.

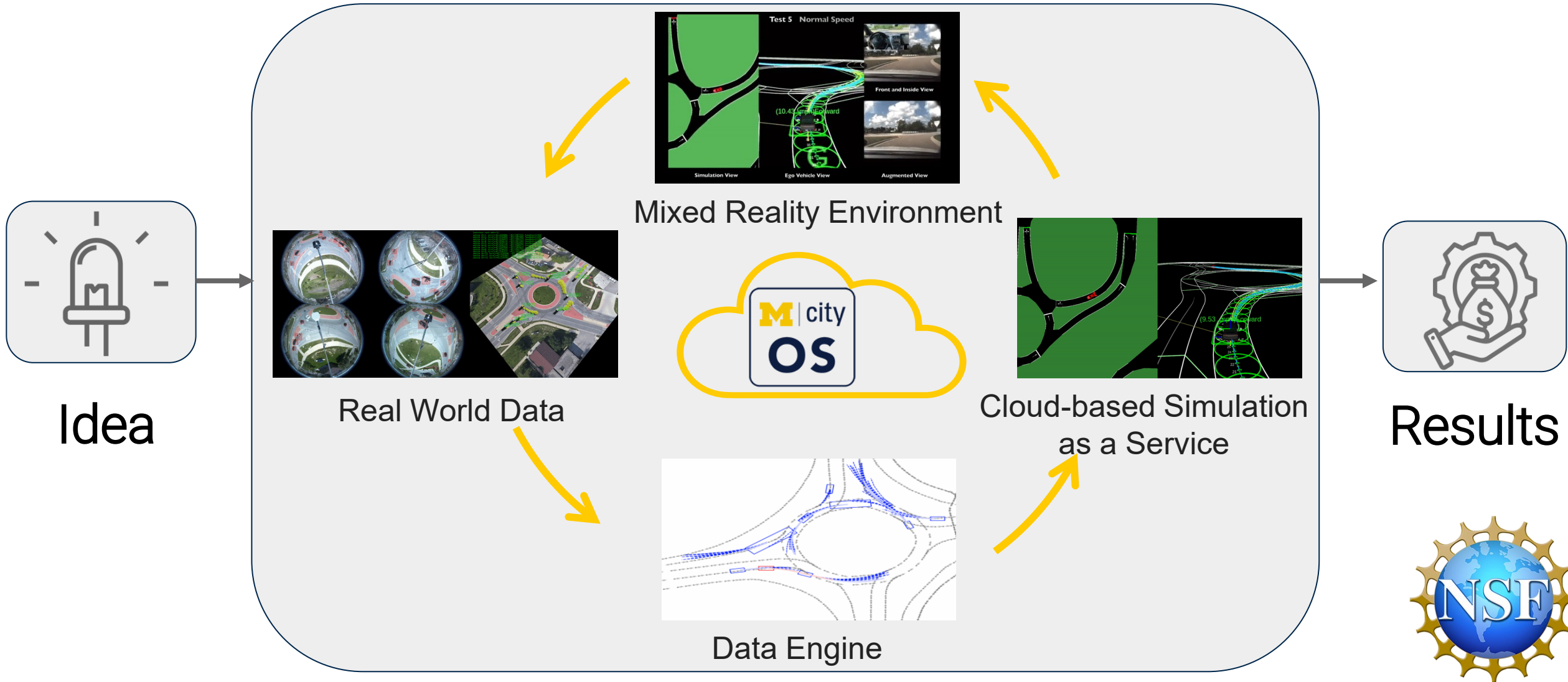
Mcity 2.0 is funded by NSF from 2022-2026. It develops digital infrastructure that builds on the physical infrastructure and enables remote access nationwide.

Mcity 1.0 (2015-21)

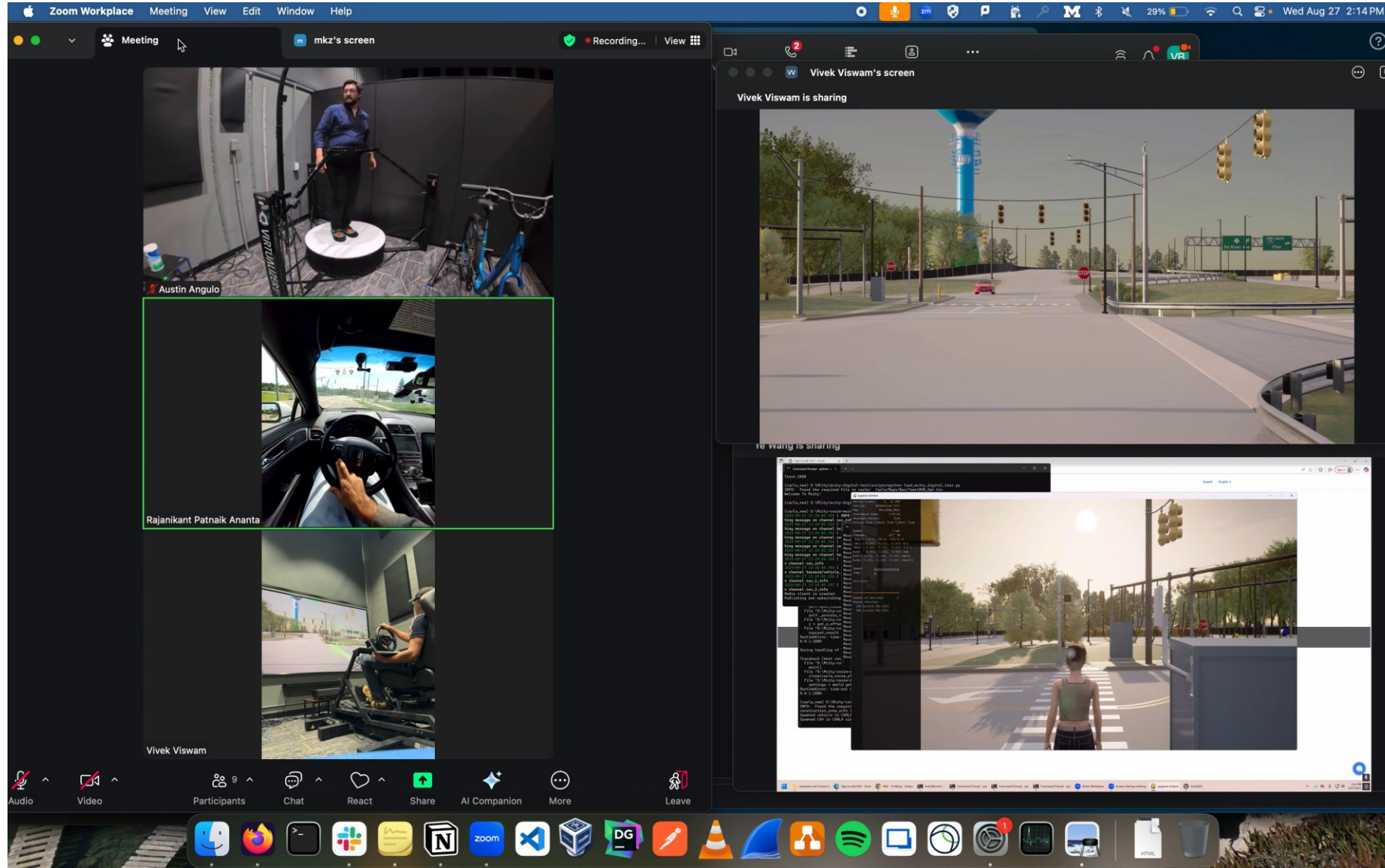
World First Testbed for
Autonomous Vehicles



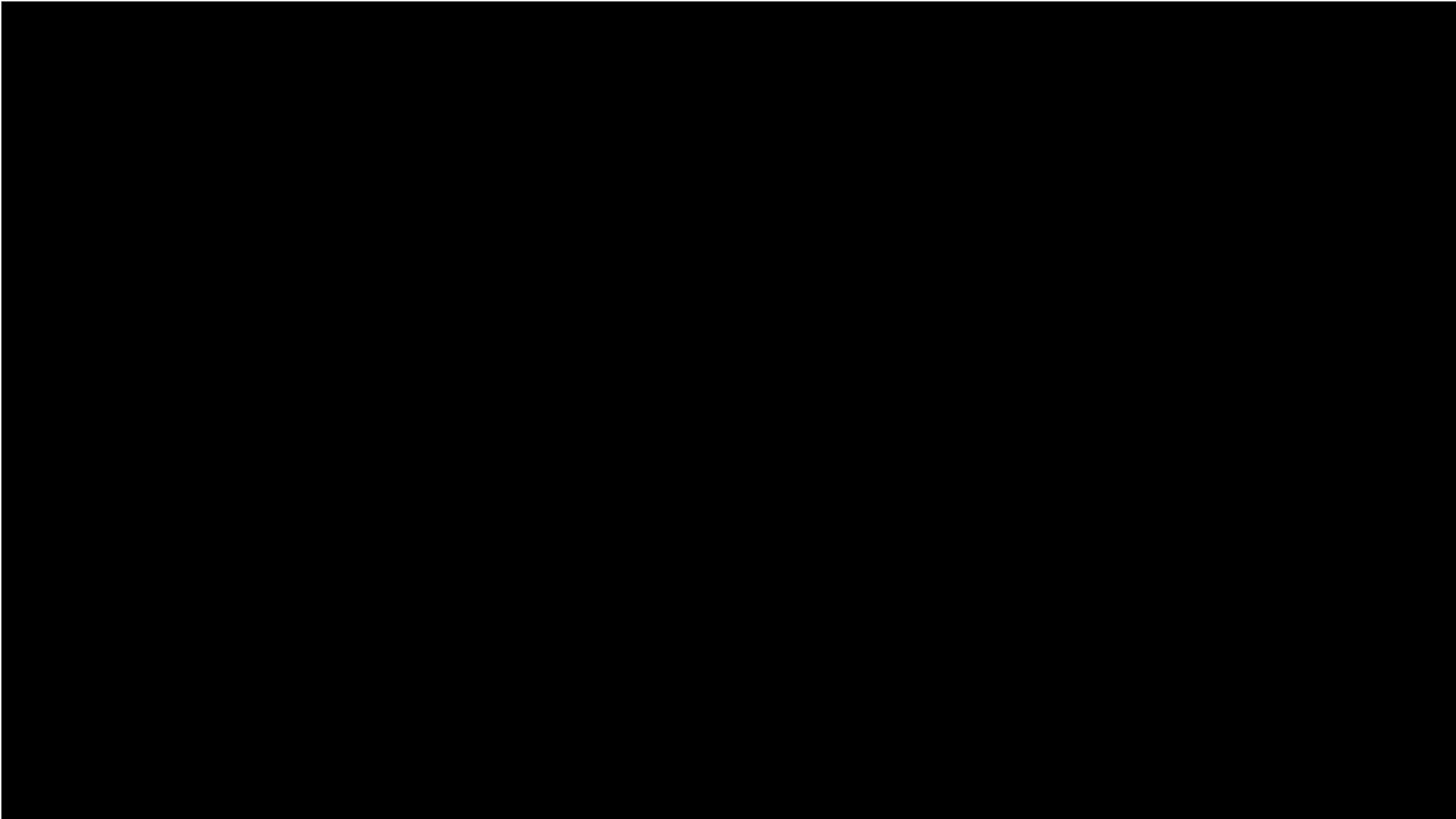
McCity 2.0: Remotely Accessible Digital Infrastructure



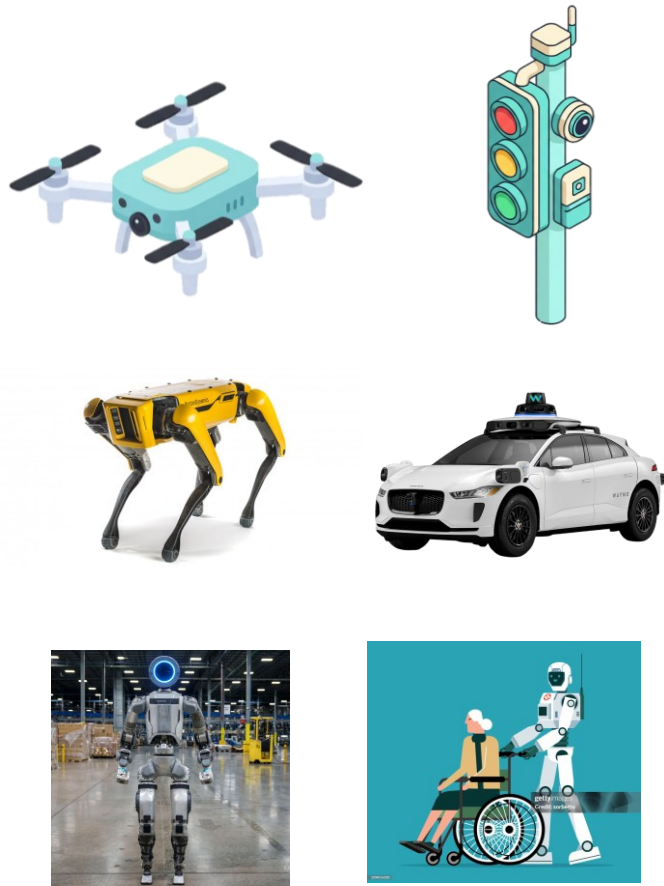
Remote testing from U of Buffalo using Mcity 2.0



Commercial use of Mcity 2.0 with Fort Robotics (August 2026)



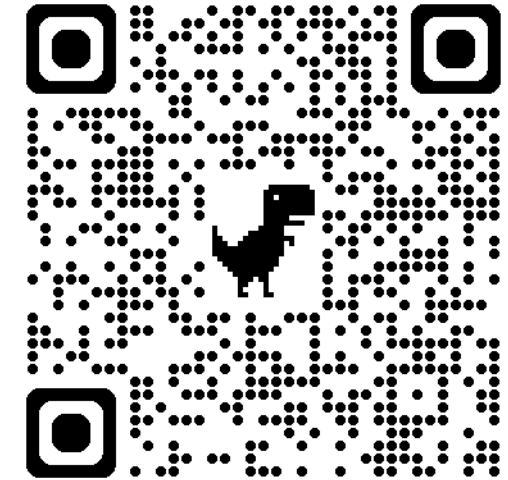
McCityX: An Urban Testbed for Physical AI



We envision a future city with physical AI serving human society. Mcity will be a physical and mixed reality environment to safely test these systems.

Paper and Source Code

- NADE Paper (2023): Dense reinforcement learning for safety validation of autonomous vehicles. *Nature*, 615(7953), 620-627.
- CoR Paper (2024): Curse of rarity for autonomous vehicles, *Nature Communications*, 15(1), 4808.
- AV Perspective Paper (2026): Two decades of safety-scalability dilemma for autonomous vehicles, *Nature*, accepted.
- TeraSim Paper (2025):
<https://arxiv.org/abs/2503.03629>



TeraSim Github Repo:

<https://github.com/mcity/TeraSim>

All other source codes are
available at:

<https://github.com/michigan-traffic-lab>

Acknowledgements



National Science Foundation
(CMMI #2223517)



USDOT Automated Driving System
Demonstration Grant
(#693JJ319NF00001)



USDOT Region 5 University
Transportation Center: Center for
Connected and Automated Transportation
(CCAT) of the University of
Michigan (#69A3551747105)



UNIVERSITY OF MICHIGAN

LEADING THE MOBILITY
TRANSFORMATION

